VideoRoPE: What Makes for Good Video Rotary Position Embedding?

UNKNOWN

01 Our Main Result: VideoRoPE is SOTA and More Robust at Long Contexts

- We achieve SOTA on LongVideoBench, MLVU, Video-MME at 8k–64k.
- Gains vs M-RoPE at 64k: +2.91 (LVB), +4.46 (MLVU), +1.66 (MME).
- We improve V-NIAH/V-NIAH-D by 12.44% and reduce hallucinations (+29.5% temporal, +18.0% object-relation).

02 How We Fix It: LTA + DL + ATS (in brief)

- We find naïve RoPEs misallocate temporal frequency and break spatial symmetry.
- We use Low-freq Temporal Allocation, a Diagonal Layout, and Adjustable Temporal Spacing.
- We prevent periodic collisions and align temporal/text scales.

03 Evaluation: Robustness and Key Insights

- We stay robust under periodic distractors (V-NIAH-D); M-RoPE drops.
- We see consistent gains to 64k; ablations: LTA and ATS contribute most.
- Takeaway: allocate lower freq to time + ATS = stable long-range reasoning.

Thank You

Questions & Discussion